Exact Solutions and Semiclassical Probes in Higher-Dimensional Gravity

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We construct two exact, static solutions of the higher-dimensional Einstein equations sourced by anisotropic stress–energy and analyze their classical and semiclassical properties. The resulting geometries describe extended, defect-like configurations with distinct global behavior: one is non-asymptotically flat, while the other contains a curvature singularity whose influence on semiclassical observables can be effectively suppressed within a controlled near-core expansion.

Classical phenomenology is investigated through effective gravitational potentials, null geodesics, and gravitational lensing. We find either saturated or short-ranged deflection profiles that differ qualitatively from the Schwarzschild case and depend sensitively on the number of spatial dimensions. The associated effective energy density and pressure exhibit a pronounced dimension-dependent structure, signaling a dimension-driven crossover in the stress–energy description.

As a semiclassical probe, we compute massless scalar two-point functions in a conformal vacuum, obtaining a dimension-dependent spectral behavior near the defect core. These results identify a regime in which global geometric properties and stress anisotropy, rather than local curvature alone, govern both classical and semiclassical observables.

Keywords: higher-dimensional gravity, topological defects, gravitational lensing, semiclassical gravity, quantum fields in curved spacetime

INTRODUCTION

The study of topological defects remains a cornerstone of modern cosmology and high-energy physics. In the early Universe, spontaneous symmetry breaking at various energy scales is expected to generate relic objects such as global monopoles and cosmic strings, whose gravitational fields correspond to localized distortions of spacetime geometry and stress–energy (Refs. 1 and 2). These configurations are not merely mathematical curiosities: they may leave observable imprints on large-scale structure, gravitational lensing, and the propagation of fields in curved backgrounds (Refs. 3 and 4). In higher-dimensional gravity, such relics provide a useful arena for probing the geometric and dynamical consequences of extra dimensions, as motivated by string-inspired and brane-world scenarios (Refs. 5–7).

The classical properties of topological defects have been extensively studied, with the Barriola–Vilenkin solution providing a paradigmatic example of a global monopole spacetime (Ref. 3). Nevertheless, several challenges persist when extending these analyses beyond four dimensions and toward semiclassical regimes. In particular, stress–energy tensors that violate energy-conditions, anisotropic pressures, and nontrivial global structure arise naturally in higher-dimensional settings, complicating both the physical interpretation of solutions and their semiclassical stability (Ref. 8).

From the semiclassical perspective, understanding quantum field behavior on defect-like backgrounds requires a careful treatment of correlation functions and vacuum states. Standard approaches to renormalization of stress–energy expectation values rely on local counterterms, point-splitting techniques, and analytic regularization methods (Refs. 9–11). In spacetimes with nontrivial topology or conical structure, quantum fields may exhibit enhanced infrared sensitivity and vacuum polarization effects, as has been extensively studied in lower-dimensional and defect geometries (Refs. 12–14).

In this work, we present two new classes of exact, static solutions to the $(n+1)$-dimensional Einstein field equations. Both solutions are characterized by a vanishing Ricci scalar and stress–energy tensors that violate the weak energy condition, making them natural candidates for effective descriptions of anisotropic, defect-like configurations in higher dimensions. The first solution is non–asymptotically flat in the weak field limit, while the second contains a naked singularity whose physical effects can be regulated at the level of observables. We emphasize that these geometries are treated here as effective spacetimes, suitable for probing classical and semiclassical phenomena rather than as fundamental microscopic descriptions, in the spirit of gravitational EFT (Refs. 6 and 15).

We investigate the classical phenomenology of these solutions through effective gravitational potentials and null geodesics. In particular, we analyze gravitational lensing and parametrize how deflection angles depend on the number of spacetime dimensions, identifying qualitative departures from four-dimensional behavior (Refs. 4, 16, and 17). We also examine the dimensional dependence of the effective pressure, which undergoes a sign change at a critical dimension, suggesting a dimension-driven transition in the space of solutions.

As a semiclassical probe, we study a quantized scalar field propagating on these backgrounds and compute two-point correlation functions in a conformal vacuum. Divergences are handled using the coincidence limit of Wightman functions within a controlled near-core expansion, following standard methods in quantum field theory in curved spacetime (Refs. 8 and 9). The use of conformal techniques and scaling arguments further enables contact with ideas from critical phenomena and conformal field theory (Refs. 18–20). This semiclassical analysis is intended as a diagnostic of vacuum response and quantum fluctuations in the presence of energy-condition violation. A systematic treatment of renormalization and backreaction effects is left for future work.

This paper is organized as follows. In Sec. I, we introduce a unified metric ansatz and derive two families of exact solutions to the higher-dimensional Einstein equations, discussing their symmetries and conformal properties. In Sec. II, we analyze the classical phenomenology, including null geodesics, constraints related to cosmic censorship, and higher-dimensional effects on gravitational lensing. In Sec. III, we present the semiclassical analysis, computing scalar field correlators in a conformal vacuum as probes of quantum behavior on these backgrounds. Finally, in Sec. IV, we discuss implications for higher-dimensional gravity and outline directions for future work, including connections to string-inspired scenarios and semiclassical backreaction.

I. NEWTONIAN SOLUTION

Given the Newtonian metric

$$\mathrm{d}s^2 = -\Psi(t,\mathbf{x})\,\mathrm{d}t^2 + \Phi(t,\mathbf{x})\delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j \ , \tag{1}$$

in $n+1$ spacetime dimensions, is straightforward to calculate the Christoffel symbols, eqs. (A1), and the Ricci associated quantities, eqs. (A2)-(A3).

Furthermore, for a compact-object we assume the metric potentials to be isotropic and static, eqs. (A4)-(A6).

Anisotropies

To trace how anisotropies origin even when the ansatz is isotropic we look at the ij-component of the Ricci

$$R_{ij} = \mathcal{A}(r)\delta_{ij} + \mathcal{B}(r)\frac{x^i x^j}{r^2}\,, \quad r := |\mathbf{x}|\,, \tag{2}$$

where the radial functions are defined in the Appendix A, eqs. (A4). Thus, the source of geometrical anisotropy is the second function.

Bernoulli constraints

In order to give a geometrical ansatz to the metric potential Φ based on symmetry requirements, we define a second set of radial functions in the Appendix A, eqs. (A5).

This approach results in a particularly elegant way of showing that constraint functions $A(r)$ and $B(r)$ (second pair) capture the "linear" curvature profile (the second derivatives), while the shift to the full Ricci tensor is purely an adjustment of the self-interaction of the scalar field gradient.

The restrictions give a non-linear second order differential equation that can be turn into a Bernoulli equation for Φ. The general solution contains two arbitrary constants which will show how the global properties of the geometry affect the gravitational constraints.

Einstein field equations

Let us consider the Einstein field equations (EFE) as defining, for a given spacetime geometry, the effective stress–energy tensor required to support it. For the static and isotropic metrics considered in this work, the most general stress–energy tensor compatible with the symmetries is necessarily anisotropic. Rather than postulating a specific matter model, we adopt a geometry-first approach: the stress–energy tensor is determined a posteriori by substituting the metric into the Einstein tensor.

The Ricci tensor simplifies by fixing $\Psi = 1$, as its time-time component vanishes. The EFE are reduced to the system

$$\frac{1}{2}R(r) - \Lambda = M_{\mathsf{P}}^{-2}\,T_{00}\,, \tag{3a}$$

$$\mathcal{C}(r)\delta_{ij} + \mathcal{B}(r)\frac{x^i x^j}{r^2} = M_{\mathsf{P}}^{-2}\,T_{ij}\,, \tag{3b}$$

$$\mathcal{C}(r) := \left(\mathcal{A}(r) - \frac{1}{2}\Phi(r)R(r) + \Lambda\Phi(r)\right), \tag{3c}$$

where $M_{\mathsf{P}} := (8\pi G_{\mathsf{N}})^{-1/2}$ is the Plank mass in terms of the Newtonian gravitational constant.

From eq. (3a) the energy density is completely determined by the Ricci scalar and the Cosmological constant. On the other hand, from the spatial components, eqs. (3b), we obtain a set of non-linear second order differential equations, where the radial functions are defined in the Appendix A, eqs. (A4)-(A5).

To make the anisotropic structure of the effective source explicit, we decompose the spatial stress tensor into an isotropic and a traceless part. Writing the stress–energy components as

$$T_{00} := \rho\,, \tag{4a}$$

$$T_{ij} := P\delta_{ij} + \tau_{ij}\,, \quad \tau^i{}_i = 0\,, \tag{4b}$$

where the indexes for the tensor τ_{ij} are raised by the Kronecker delta. The scalar functions $\rho(r)$ and $P(r)$ represent the energy density and isotropic pressure respectively, while τ_{ij} encodes the anisotropic shear stress required to support the geometry.

This decomposition is purely kinematical and follows from the symmetry of the stress–energy tensor. No equation of state or microscopic matter model is assumed.

I.A. Exotic void

From the Bernoulli constraints, let us set $B = 0$ in eqs. (A5) to decrease the anisotropy sourced by the geometry. By solving the constraint on $\Phi(r)$, the general solution contains two constants of integration, namely a and b.

We can set the second constant so that we obtain a Minkowski spacetime in the limit when $r \to 0$. While the other constant can be set to $a := r_c^2\,[L^2]$, where r_c is the core radius. We call this choice **normalized units** in the following. We interpret the core as a sphere containing the effective volume of the exotic model. Thus, we denote the metric potential as

$$ {}^{(1)}\Phi(s) := \left(1 + ns^2\right)^{\beta} \,, \tag{5a} $$

$$ \beta := 2/n \,, \quad s := r/r_c \,. \tag{5b} $$

Let us take $\Lambda = 0$ in eqs. (3). The effective matter content is given by

$$ {}^{(1)}\rho = -\frac{2M_{\mathsf{P}}^2(n-1)}{r_c^2}\left[\frac{n}{u} - \frac{(n+2)s^2}{u^2}\right], \tag{6a} $$

$$ {}^{(1)}P = \frac{2M_{\mathsf{P}}^2}{r_c^2 u}\Bigg[2(1-n) + \frac{s^2}{u} + n(n-1)u^{\beta} + (n-1)(n+2)s^2u^{\beta-1}\Bigg], \tag{6b} $$

$$ {}^{(1)}\tau_{ij} = {}^{(1)}\Pi(s)\left(\frac{x^i x^j}{r^2} - \frac{1}{n}\delta_{ij}\right), \tag{6c} $$

$$ {}^{(1)}\Pi(s) := \frac{4M_{\mathsf{P}}^2}{r_c^2}\frac{(n+1)(n-2)s^2}{r_c^2\,u^2}\,, \tag{6d} $$

$$ u(s) := \left(1 + ns^2\right). \tag{6e} $$

Equations (6) describe the effective stress–energy tensor required to support the void geometry. The matter content is intrinsically anisotropic and fully determined by the normalized variable s. The energy density ρ is sourced by the non–vanishing Ricci scalar and exhibits a strong dependence on the spacetime dimension n. The isotropic pressure P contains contributions independent of ρ and power-law terms in u, indicating that the source cannot be modeled as a perfect fluid. These contributions dominate near the core and encode the dimensional dependence of the effective equation of state. The anisotropic stress tensor τ_{ij}, governed by $\Pi(s)$, provides the shear required to sustain the geometry and becomes increasingly relevant with dimension.

Energy conditions

Several generic features follow directly from eqs. (6). First, the energy density is generically negative for $n > 1$ over a wide range of radii. Near the core ($s \to 0$), one finds

$$ \rho \to -\frac{2M_{\mathsf{P}}^2(n-1)n}{r_c^2} < 0 \,, \tag{7} $$

indicating a violation of the weak energy condition (WEC). Also the null energy condition (NEC) is violated near the core as the energy density dominates.

Second, although the void violates the NEC in the interior, the violation is localized: in the far-field region the positive pressure and anisotropic shear dominate over the negative energy density, so null congruences satisfy the NEC asymptotically.

Third, the anisotropic stress τ_{ij} is controlled by the function $\Pi(s) \propto (n-2)s^2/u^2$, which vanishes at the core and grows away from it. This term encodes directional stresses characteristic of defect-like configurations and becomes nontrivial for $n \neq 2$. Its traceless form ensures that anisotropy, rather than isotropic pressure alone, plays a key role in supporting the geometry.

Taken together, these properties show that the void solution is sustained by exotic, anisotropic matter that violates the weak energy condition, a feature commonly encountered in effective descriptions of topological defects and regularized void-like geometries. The energy-condition violation is not pathological in this context but rather underlies the absence of a Newtonian attractive potential and the emergence of a repulsive core that renders the void effectively impenetrable to null and time-like probes, as is shown in Sec. II.

Fig. 1 illustrates the radial profile of the effective energy density and pressure. These plots are included to visualize the localization and sign structure of the stress–energy tensor and to indicate the regions where classical energy conditions are violated

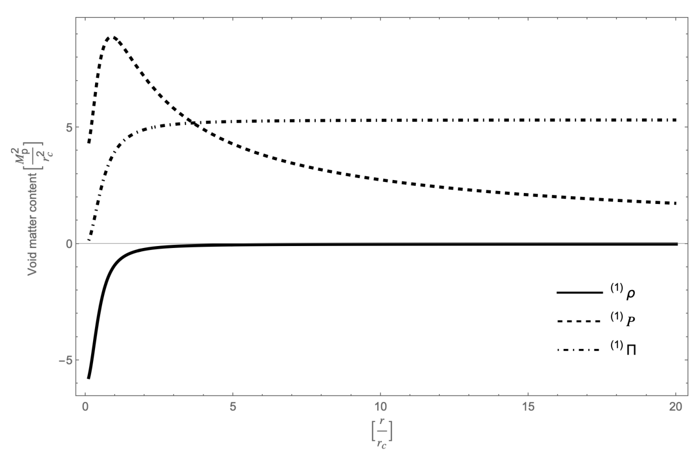

FIG. 1: The void exhibits decaying pressure and energy density but a long-ranged anisotropic stress, highlighting its defect-like gravitational character.

Curvature

Following our geometric approach, we write the Ricci scalar in normalized units as

$$^{(1)}R(s) = \frac{4(1-n)}{r_c^2}\left[\frac{n}{(1+ns^2)} - \frac{(n+2)s^2}{(1+ns^2)^2}\right]. \tag{8}$$

The Ricci scalar, eq. (8), indeed reveals the dual nature of the defect through its two constituent terms for dimensions n bigger than one. Where the first term establishes the monopole behavior of the source, providing a localized peak of negative curvature at the origin that scales with the dimension. Whereas the second term acts as a geometric regulator, vanishing at the origin and peaking near the core radius, it facilitates the transition toward asymptotic flatness. Together, they define the "void" profile as a non-singular, repulsive gravitational core that is topologically distinct from the surrounding flat spacetime.

For instance, by taking the limit $r \to 0$, the curvature tends to a constant value

$$^{(1)}R(r \to 0) = -\frac{4n(n-1)}{r_c^2}, \tag{9}$$

which is inversely proportional to the squared radius of the void's core. Meaning that a **tight core** with small r_c creates a highly curved "spike" of negative curvature. Conversely, a **wide core** having a large r_c creates a shallow, nearly flat "bubble."

We see that, unlike standard decaying potentials, the metric potential (5) describes a spacetime in which the spatial volume grows with the radial coordinate. This behavior can be interpreted as a form of vacuum polarization, in which anisotropic vacuum tension, rather than a mass monopole, drives a geometric expansion of spatial sections.

I.B. Anisotropic shell

As a subtle consequence of the Bernoulli constraints we find that, by increasing the stress anisotropy, with the condition $A = 0$ imposed in eqs. (A5), the metric potential Φ shows an inverted behavior to the void, attracting rather than expanding.

To see this let us take the general solution to the Bernoulli constraint and set properly the two constants of integration, a and b, to obtain a Minkowski limit, while also choosing normalized units. We denote the metric potential by

$$^{(2)}\Phi(s) := \left(1 + s^{-\alpha}\right)^{\beta}, \tag{10a}$$

$$\alpha := 2n - 4, \quad \beta = 2/n, \quad n \neq 2, \tag{10b}$$

which matches flatness when $r \to \infty$. In this case, the normalized units are defined by setting $a := r_c^{\alpha}[L]^{\alpha}$ in the first integration constant. In contrast with the void, the shell metric potential increases as $s \to 0$ producing an effective potential well that focuses null geodesics toward the core. This behavior signals an attractive gravitational response, despite the absence of a mass monopole, and is sourced by the localized anisotropic stress rather than by positive energy density.

Let us study the effective matter content as before, by taking $\Lambda = 0$ in eqs. (3), we obtain

$$^{(2)}\rho = \frac{2M_{\mathsf{P}}^2}{r_c^2}\frac{(n-1)(n-2)^2}{n^2}\frac{2 - ns^{\alpha}}{s^2(1+s^{\alpha})^2}, \tag{11a}$$

$$^{(2)}P = \frac{2M_{\mathsf{P}}^2}{r_c^2}\frac{(n-2)^2}{n^2}\left(\frac{4}{s^2(1+s^{\alpha})^2} - \frac{(n-1)(2-ns^{\alpha})}{s^{2+\alpha\beta}(1+s^{\alpha})^{2-\beta}}\right), \tag{11b}$$

$$^{(2)}\tau_{ij} = {}^{(2)}\Pi(s)\left(\frac{x^i x^j}{r^2} - \frac{1}{n}\delta_{ij}\right), \tag{11c}$$

$$^{(2)}\Pi(s) := -\frac{4M_{\mathsf{P}}^2}{r_c^2}\frac{(n-2)^2}{n^2}\frac{2 + n(n-1)s^{\alpha}}{s^2(1+s^{\alpha})^2}. \tag{11d}$$

As with the void, the structure reflects the localized, defect-like nature of the shell rather than a conventional perfect fluid.

Energy conditions

The energy density is singular at the core, scaling as $\rho \sim s^{-2}$, which signals the presence of a genuine curvature-supported defect. Its sign is controlled by the factor $2 - ns^{\alpha}$, in the region $s^{\alpha} < 2/n$, the energy density is positive, while it becomes negative beyond this radius. This sign change coincides with the geometric transition identified by the vanishing of the Ricci scalar and indicates that the shell interpolates between a positive-energy interior and an exotic exterior region.

The isotropic pressure P exhibits a more intricate radial dependence, with competing terms that scale differently with s. Near the core, the leading contribution is negative and divergent, whereas at larger radii the pressure becomes positive and zero asymptotically due to the dominance of the second term. Consequently, the combinations $\rho + P$ generically become negative in part of the spacetime, implying violations of the null and weak energy conditions in the exterior region of the shell, as plotted in fig. 2.

The anisotropic stress τ_{ij} plays a central role in supporting the configuration. The function $\Pi(s)$ is negative definite and diverges as s^{-2} at the core, indicating strong directional stresses. The traceless structure of τ_{ij} further emphasizes that the shell is not supported by isotropic pressure alone, but by shear-like stresses characteristic of extended topological defects.

Overall, the shell solution is sustained by localized, anisotropic exotic matter that violates standard energy conditions in a controlled manner. As plotted in fig. 2, both the anisotropic stress and the (negative) energy density are sharply localized near the core and decay rapidly in the weak-field region, reflecting the asymptotic flatness of the geometry despite strong near-core violations of the energy conditions.

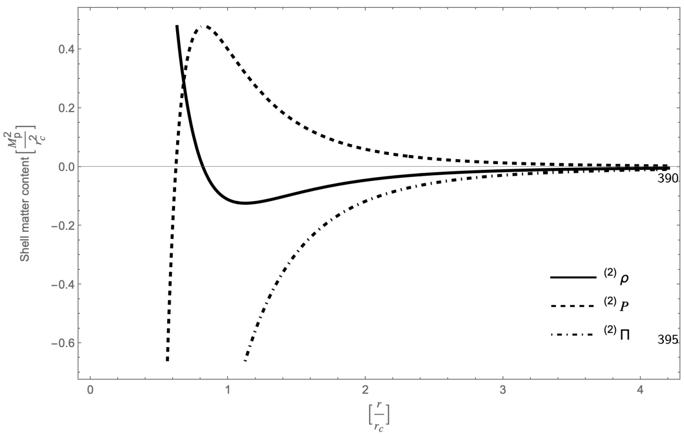

FIG. 2: The shell exhibits divergent stresses and negative energy density near the core, while all effective sources decay at large radius, consistent with asymptotic flatness.

Curvature

Let us examine the Ricci scalar for the shell,

$$^{(2)}R(s) = \frac{4(n-1)(n-2)^2}{r_c^2\, n^2}\,\frac{2 - n s^\alpha}{s^2 (1+s^\alpha)^2}\,. \tag{12}$$

The scalar characterizes the intrinsic curvature of the shell geometry as a function of the dimensionless radius $s = r/r_c$.

Near the core ($s \to 0$), the curvature diverges as $R \sim s^{-2}$, signaling a curvature singularity localized at the origin. This divergence is independent of α and reflects the presence of a defect-like core rather than a conventional matter distribution.

In the region defined by $s^\alpha < 2/n$, the geometry possesses positive curvature, corresponding to a locally spherical geometry. At the critical radius

$$s_0 = (2/n)^{1/\alpha}\,, \tag{13}$$

the scalar vanishes, marking a geometric transition. Beyond this radius ($s^\alpha > 2/n$), the curvature becomes negative ($R < 0$), indicating a hyperbolic-like exterior region. Finally, in the limit $s \to \infty$, the curvature decays as $R \sim s^{-(\alpha+2)}$, ensuring the metric satisfies the condition of asymptotic flatness at large distances. This behavior confirms that the shell geometry represents a localized curvature defect embedded in an asymptotically flat background.

Is worth to notice that in dimension $n = 1$, or (1+1) gravity, the two solutions represent the same spacetime as there is only one dimension and thus anisotropy cannot be generated.

Symmetries

From the static and isotropic conditions is clear that the spacetime metric

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \Phi(r)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\,, \tag{14}$$

admits time translation and rotational invariance.

Nevertheless, because of the radial dependence we should not expect spatial translations to hold. As one can see from the Killing eq. (A7c), these are not symmetries of the transverse metric.

However, a hidden conformal translation symmetry does appear. We see this by looking at the eq. (A7c), from which we obtain the **Conformal Killing Equation** (CKE)

$$\mathcal{L}_\xi g_{ij} = 2\psi\, g_{ij}\,, \tag{15}$$

in the special case when

$$\xi^0 = 0\,, \quad \xi_i = C_i\ \Phi(r)\,, \tag{16a}$$

$$\psi(\mathbf{x}) = (\mathbf{C}\cdot\mathbf{x})\frac{1}{2r}\frac{\Phi'}{\Phi}\,. \tag{16b}$$

The semiclassical impact of this result is that the transverse space is conformally flat, which means that we can use the quantization techniques applied in Minkowski, after defining a conformal frame.

Singularities

After deriving the solutions and looking at the conformal Killing equations, it is worth noticing that the spacetime backgrounds (14) can be regarded as **ultra-static**, corresponding to the choice $\Psi = 1$ in the Newtonian ansatz of eq. (1). In this frame the time coordinate coincides with proper time for static observers, and all nontrivial geometric information is encoded in the spatial conformal factor $\Phi(r)$.

To assess the physical regularity of the spacetime, it is therefore necessary to go beyond curvature diagnostics and examine geodesic completeness. For radial time-like geodesics one finds

$$\dot{r}^2 = \frac{E^2 - 1}{\Phi(r)}\,, \tag{17}$$

so that the proper time required for an observer starting at radius r_0 to reach the core is

$$\Delta\tau = \int_0^{r_0} \sqrt{\Phi(r)}\,\mathrm{d}r\,, \tag{18}$$

where $\sqrt{\Phi(r)} \sim r^{(2n-n^2)/2}$, for the shell (10) if $n > 2$.

Then, this integral converges whenever the exponent is bigger than -1, as is the case for all $n \geq 3$. Consequently, time-like geodesics reach $r = 0$ in finite proper time, establishing geodesic incompleteness and a nakedness symptom for the shell's singularity.

Ultrastatic frame

While the analysis in Sec. II focuses on classical phenomenology, such as effective potentials, null and timelike geodesics, and gravitational lensing, it is important to clarify the role of the spacetime frame employed in each regime. Classical observables of this kind are invariant under coordinate and conformal redefinitions of the metric and are therefore most transparently studied in the ultrastatic frame, where the causal structure and topological features of the solutions are manifest. In this frame, geodesic completeness, effective forces, and asymptotic deflection angles can be analyzed without ambiguity.

Conversely, the semiclassical effects explored in Sec. III, including quantized field correlators, effective power spectra, and vacuum fluctuations, are inherently sensitive to redshift, mode normalization, and the definition of positive frequency. Following standard practice in quantum field theory on curved spacetimes, and in close analogy with treatments of quantum fluctuations in quasi–de Sitter inflation, we therefore perform the semiclassical analysis in a conformally related frame.

The conformal frame provides a technically natural setting for defining vacuum states and computing Wightman functions, particularly for massless or conformally coupled fields, and facilitates controlled near-core expansions within an effective field theory framework. While the conformally related metric is not dynamically equivalent to the original spacetime at the level of the Einstein equations, it preserves the causal structure and serves here as an auxiliary geometry for probing semiclassical response rather than as a fundamental gravitational background.

Geometric interpretation

We emphasize that the resulting gravity sources should be interpreted as effective descriptions of anisotropic stresses induced by extended or defect-like configurations, rather than as a fundamental classical fluid. In this sense, the appearance of energy-condition violations is not assumed but emerges as a property of the effective source required to sustain the geometry.

Such an interpretation is consistent with an effective field theory treatment of gravity, in which the metric encodes coarse-grained geometric effects while the corresponding stress–energy tensor captures nonlocal, topological, or quantum-induced contributions. The validity of this description is restricted to scales larger than the characteristic core radius r_c, beyond which a microscopic completion would be required.

II. GRAVITATIONAL CONSTRAINTS

In this section we present the two exotic models, study their gravitational properties via the metric (14) and then apply these results to their classical phenomenology.

Effective potential

To calculate the energy conservation law for a test particle in this metric we can use the conserved quantities of eqs. (A9) along with the normalization condition of the $(1+n)$-velocity, $u^\mu u_\mu = \epsilon$,

$$\frac{\dot{r}^2}{2} + V_{\text{eff}}(r) = \frac{E^2}{\Phi(r)}\,, \tag{19a}$$

$$V_{\text{eff}}(r) := \frac{L^2}{2\Phi^2(r)r^2} - \frac{\epsilon}{2\Phi(r)}\,. \tag{19b}$$

These expressions show that the total energy of the particle is diluted or enhanced according to the behavior of the metric potential $\Phi(r)$.

Let us fix $n = 3$. The effective potentials in **normalized units**, for eqs. (5) & (10), are

$$^{(1)}V_{\text{eff}}\,(s) := \frac{\chi^2}{2s^2\left(3s^2+1\right)^3}\,, \tag{20a}$$

$$^{(2)}V_{\text{eff}}\,(s) := \frac{s^4\chi^2}{2\left(s^2+1\right)^3}\,, \tag{20b}$$

$$\chi := L/r_c\,, \quad s := r/r_c\,.$$

Hence, our normalized units measure the photon's proximity to the core in units of the core's own size. Also, the ratio χ can be viewed as a dimensionless **reduced angular momentum**.

This result proves that the lensing and geodesic behavior are self-similar. A relic with a core the size of a galaxy and a relic the size of a proton will deflect light in the exact same way, provided the ratio of the impact parameter to the core size is the same. We plotted the graphs of this potentials in fig. 3 and 4.

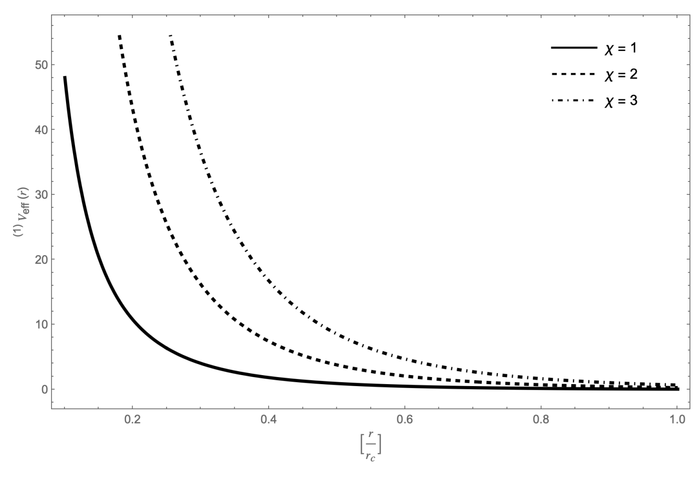

FIG. 3: The Void potential is absolutely impenetrable.

We interpret the centrifugal barrier exhibit by the **void effective potential** (fig. 3) as a consequence of the spacetime being locally flat at the origin. In this case, curvature effects do not counteract the centrifugal contribution to the radial motion of null geodesics. As a result, trajectories with nonzero angular momentum are

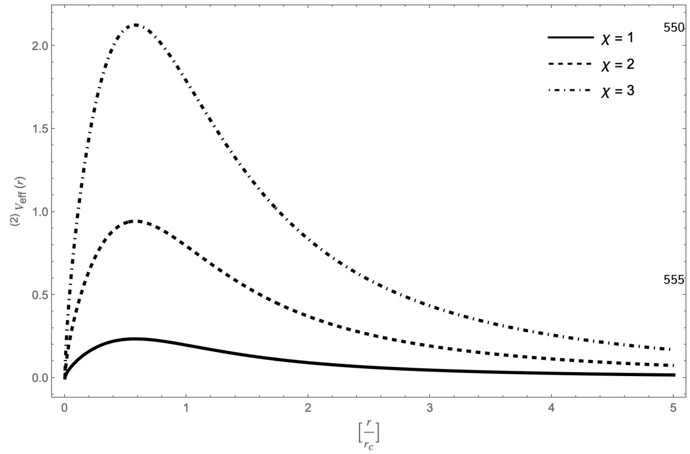

FIG. 4: The Shell anisotropy overwhelms the centrifugal barrier.

repelled from the core, and null geodesics do not reach $r = 0$.

This behavior implies that the central region is classically inaccessible to test particles, rendering any possible singular structure at the origin operationally irrelevant at the level of geodesic probes. In this sense, the void acts as an effectively "safe" defect. Although the geometry may admit a formal singular limit, physical observables associated with null propagation are shielded by the centrifugal barrier.

Notably, this protection persists even in the formal limit $r_c \to 0$, indicating that the core remains unprobed within the classical regime of validity of the solution.

In contrast to the void, the shell solution exhibits a genuine naked singularity at $s = 0$ ($r = 0$ in eq. (10), as shown by eq. (18)). This can be seen directly from the structure of the **shell effective potential**, fig 4. While centrifugal barriers typically prevent light rays from accessing the central region in regular or horizon-shielded spacetimes, the shell potential does not diverge sufficiently to provide such protection. As a result, null geodesics with sufficiently large conserved energy are able to overcome the potential barrier and reach $s = 0$ in finite affine parameter.

The absence of a horizon or turning point separating the singularity from the exterior implies that the singularity is causally accessible to asymptotic observers, thereby satisfying the standard criterion for nakedness. This behavior contrasts sharply with black hole geometries, where the effective potential and horizon structure prevent direct geodesic access to the singular region.

It should be emphasized, however, that the existence of a naked singularity at the classical level does not preclude the possibility of semiclassical screening, since quantum observables probe the regulated geometry defined by the shell.

Lensing

From the angular momentum and the conservation law, eqs. (A9b)&(19), we write the orbit of a photon as

$$\frac{\mathrm{d}\phi}{\mathrm{d}r} = \frac{1}{r\sqrt{2\Phi(r)\left(\frac{r}{\mathtt{b}}\right)^2 - 1}}\,, \qquad \mathtt{b} := L/E\,, \tag{21}$$

where $\epsilon = 0$.

The **deflection angle** α can be plotted in terms of the impact parameter $\mathtt{b}$, after integrating the orbit eq. (21) as

$$\alpha(\mathtt{b}) := \Delta\phi(\mathtt{b}) - \pi\,, \tag{22a}$$

$$\Delta\phi(\mathtt{b}) = 2\int_{r_0}^{\infty} \frac{\mathrm{d}r}{r\sqrt{2\Phi(r)\left(\frac{r}{\mathtt{b}}\right)^2 - 1}}\,, \tag{22b}$$

where r_0 is the turning point, which is found by solving the square root of eq. (21) equal to zero.

Photon sphere

Because the shell effective potential, eq. (20b), fig. 4, has a maximum value, there is a critical impact parameter $\mathtt{b_c}$. When the photon energy is insufficient to overcome the effective potential barrier, corresponding to impact parameters $\mathtt{b} > \mathtt{b_c}$, null geodesics reach a turning point and are scattered back to infinity. As the impact parameter is decreased toward the critical value $\mathtt{b_c}$, the turning point approaches the radius of an unstable circular null orbit. This orbit defines a photon sphere in the kinematic sense, which marks the transition between scattering and capture.

Unlike black hole spacetimes, however, the deflection angle α remains finite as $\mathtt{b} \to \mathtt{b_c}$. Although the photon sphere corresponds to an extremum of the effective potential, the absence of a long-range Newtonian tail prevents the logarithmic divergence characteristic of strong lensing in asymptotically Schwarzschild geometries. Consequently, photons do not undergo arbitrarily many windings around the shell.

By maximizing the shell effective potential, eq. (20b), we find the radius of the photon sphere to be

$$r_p = \frac{\sqrt{3}}{3} r_c\,, \tag{23}$$

and the maxima

$$V_{\mathtt{max}} = \frac{3}{2^{11/3}}\chi^2\,. \tag{24}$$

The critical impact parameter is obtained by evaluating the turning-point condition at the photon sphere, yielding

$$\mathtt{b_c} = \frac{2^{7/6}}{\sqrt{3}} r_c\,. \tag{25}$$

In the numerical evaluation of the lensing integrals in eqs. (22), we therefore restrict to $\mathtt{b} > \mathtt{b_c}$, for which null geodesics are scattered and the deflection angle is well defined.

The deflection angles are plotted in figs. 5-7, where we are using the weak field limit for Schwarzschild/Tangherlini(1+3) (see eq. (7.10.29) in Ref. 8, we have taken $G^{(D)}M = 1$).

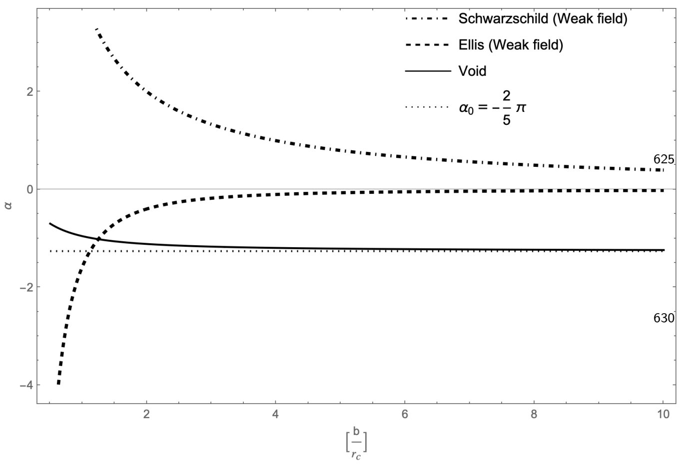

FIG. 5: The void presents a repulsive lensing stronger than the Ellis wormhole, showing a topological defect.

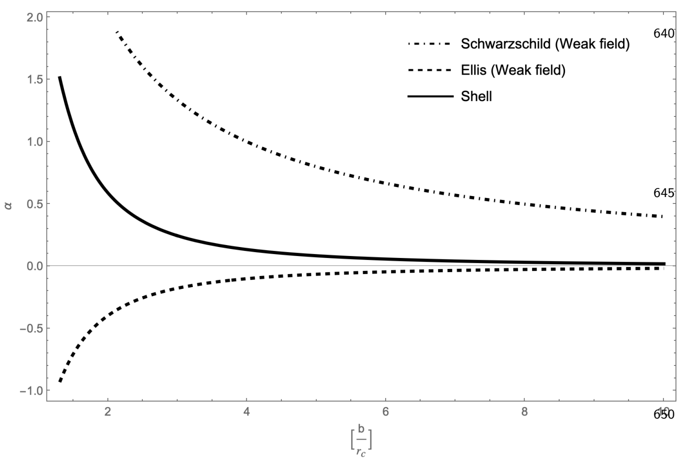

FIG. 6: The shell lensing is weaker than Schwarzschild/Tangherlini(1+3) in the weak field limit.

Conical deficit

From the numerical results, a qualitative feature of the void is that the deflection angle does not decay to zero at large impact parameter, but instead approaches a constant value. This saturation of the deflection angle indicates that the void spacetime approaches a conical geometry characterized by a deficit angle. Such behavior is well known in the context of topological defects, including global monopoles and cosmic strings, where lensing arises from global geometric properties rather than from a localized mass distribution.

To quantify the transition from the core region to the asymptotic regime, we expand the deflection angle for large impact parameter. In this limit, the geometry approaches a conical manifold, and the leading contribution to the deflection angle is a constant α_0, corresponding to the deficit angle. The first sub-leading correction encodes how rapidly the geometry interpolates between the defect core and the asymptotic cone. In Appendix B, eqs. (B2) & (B4), we show that, in the weak field limit,

$$\alpha(\mathtt{b}) = \alpha_0 + \mathcal{O}\Big([\mathtt{b}/r_c]^{-6/5}\Big)\,, \quad \alpha_0 = -\frac{2}{5}\pi\,. \tag{26}$$

The sub-leading correction decays faster than the Schwarzschild behavior $\alpha \sim \mathtt{b}^{-1}$, reflecting the absence of an effective mass monopole and of a long-range Newtonian potential. Consequently, lensing in the void geometry is governed primarily by global geometric structure rather than by local curvature, resulting in a constant asymptotic deflection supplemented by a rapidly decaying power-law correction.

This behavior closely parallels that of global monopole spacetimes, where the leading deflection is constant due to the solid deficit angle introduced by the topological charge of the defect, independent of the impact parameter in the weak field limit (Ref. 3, 4, and 17). In contrast, wormhole geometries such as the Ellis wormhole exhibit a deflection angle that decays with impact parameter, with the weak field expansion showing a leading $\propto \mathtt{b}^{-2}$ term in the asymptotic regime characteristic of massless, non-monopolar geometries (see eq. (14) in Ref. 16, we plotted this function with $a = 1$).

The presence of a constant asymptotic deflection together with a fractional power-law correction therefore provides a clear observational discriminator between void-like defects, wormholes, and compact objects, and constitutes a distinctive fingerprint of the void geometry.

Dimensional structure

In higher dimensions, the Tangherlini solution provides the canonical lensing benchmark associated with a localized mass monopole, for which the deflection angle decays as $\alpha \sim \mathtt{b}^{-(n-2)}$. In contrast, the anisotropic shell considered here does not source a monopole term in the asymptotic metric. As a consequence, its lensing behavior falls into a different universality class, characterized by a faster decay governed by higher-dimensional, non-monopolar operators. Increasing dimensionality further suppresses the shell-induced deflection relative to the Tangherlini case, as plotted in fig. 7.

However, the hierarchical behavior flips in lower-dimensional (1+1) gravity, as the localized mass monopole does not provide enough strength to be perceivable by the curvature. On the contrary, the only source of gravity measurable in this context are topological defects, such as the one provided by the shell or the void, which coincide in $n = 1$, as seen in fig. 7.

Energy violations

Even though the NEC violations are usually avoided in classical GR, they are expected in higher-dimensional

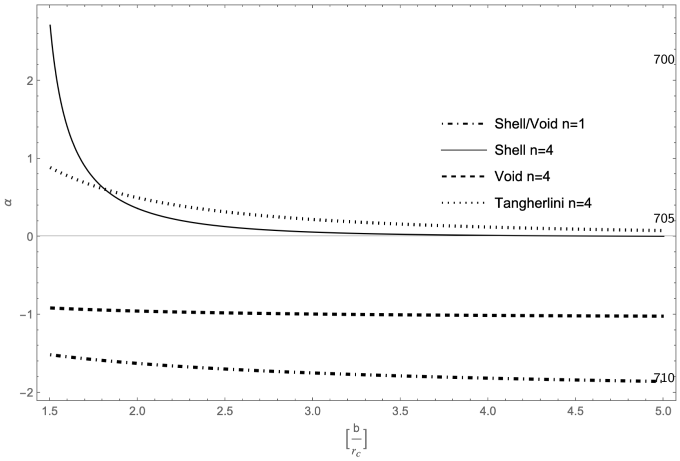

FIG. 7: The coincidence of the shell and void deflection in $n = 1$ reflects the topological character of $(1 + 1)$ dimensional gravity, while in $n = 4$ the shell and Tangherlini geometries exhibit distinct tension- and mass-sourced lensing, respectively.

defects or semi-classical gravity on the basis that quantum corrections can actually overcome the violations. On these grounds, the study of a quantized field in these backgrounds is motivated, as is done in Sec. III.

III. QUANTUM CORRELATORS

In this section, we investigate the semiclassical properties of the void and shell geometries by studying vacuum correlators of a quantized massless scalar field on the conformally related background.

Conformal frame

The conformal Killing equation (16) indicates that the spacetime metric (14) admits a conformal symmetry, suggesting the existence of a representation in which this structure is made explicit.

We therefore introduce the conformal time coordinate

$$\mathrm{d}\tau := \frac{1}{\sqrt{\Phi(r)}}\mathrm{d}t\,, \tag{27}$$

which allows the metric to be expressed as

$$\mathrm{d}s^2 = \Omega^2(x)\big(-\mathrm{d}\tau^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\big)\,, \qquad \Omega^2(x) := \Phi(r)\,. \tag{28}$$

We emphasize that eq. (28) does not represent a mere coordinate reparametrization of the original spacetime, but instead defines a conformally related metric. While the two metrics share the same causal structure, they are generally inequivalent as solutions of the Einstein field equations.

In particular, the conformal frame corresponds to choosing $\Psi = \Phi$ in the Newtonian ansatz, in contrast with the ultrastatic choice $\Psi = 1$.

In what follows, the conformal frame will be employed as an auxiliary geometry for the semiclassical analysis of quantum field correlators, where the conformal properties of massless fields play a central role.

In this conformal frame, the geometry is explicitly related to Minkowski spacetime by a local Weyl rescaling. While this transformation leaves null geodesics and the causal structure invariant, it reorganizes redshift effects and tidal distortions into the conformal factor $\Omega(x)$.

The standard results of QFT in curved space (see Ref. 9) make the quantization of a massless scalar φ straightforward for a spacetime conformal to Minkowski, as discussed in the appendix C. Using the **conformal vacuum**, the Wightman function is

$$\langle\hat{\varphi}(x)\,\hat{\varphi}(x')\rangle = \Omega^{-\frac{n-1}{2}}(x)\,\bar{G}^{+}(x,x')\,\Omega^{-\frac{n-1}{2}}(x')\,, \tag{29}$$

where $\bar{G}^{+}(x,x')$ is the Minkowski Wightman function.

The causal structure of the conformally flat spacetime is inherited from Minkowski space. Since the background is static but spatially inhomogeneous, our interest lies in how vacuum fluctuations are geometrically dressed by the conformal factor $\Omega(r)$. For this purpose, we focus on Wightman functions rather than time-ordered propagators, as they directly encode equal-time correlations and allow for the definition of effective power spectra in local orthonormal frames and in the asymptotic region. This distinction is essential in static but inhomogeneous spacetimes, where translational invariance is broken and momentum space admits only a local or asymptotic interpretation.

A textbook calculation derived in the Appendix C, eqs. (C4)-(C5), shows that the mathematical expression for the effective power-spectrum corresponds to taking the equal-time Wightman function as in eq. (29).

Localized power spectra

In order to define the effective power spectra with physical significance, we take the mathematical coincidence limit when $\mathbf{x} \to \mathbf{x}'$,

$$\mathcal{P}_{\rm loc}(r;k) := \Omega^{1-n}(r)\frac{1}{2k}\,, \quad r \ll 1. \tag{30}$$

To then state its validity only within a regulated region defined by the background itself via the core radius of both the void and the shell.

Under the assumption that the distance between two points scales as the physical scale of vacuum fluctuations, $r \sim \lambda$, we are assuming locality of UV physics, dependency of short-distance correlations only on the local geometry, as well as irrelevancy of global singularities or asymptotics at leading order.

In this limit, the void metric (10) is written as

$$^{(1)}\Phi(s) \sim 1 + n\beta s^2 + \cdots\,, \quad s = r/r_c\,, \tag{31}$$

using normalized units $s \ll 1$. Thus, the power-spectrum of the void model reduces to

$$^{(1)}\mathcal{P}_{\rm loc}(r;k) \approx \frac{1}{2k} + \frac{n\beta\sigma}{2r_c^2k^3} + \cdots\,, \quad \sigma := -\frac{n-1}{2}\,, \tag{32}$$

where the spectral index to leading order is

$$^{(1)}n_s = -(n-1)s^2 + \mathcal{O}\big(s^4\big)\,, \tag{33}$$

which is consistent with Pos-Minkowskian analysis.

On the other hand, let us expand the shell metric (10)

$$^{(2)}\Phi(s) \sim s^{-\alpha\beta} + \beta s^{-\alpha(\beta-1)} + \cdots\,, \quad s = r/r_c\,. \quad (34)$$

The power spectra near the shell's core is

$$^{(2)}\mathcal{P}_{\rm loc}(r;k) \approx \frac{1}{2} r_c^{\eta} k^{\eta-1}\,, \quad \eta := -\alpha\beta\frac{n-1}{2}\,. \quad (35)$$

Therefore, the spectral index of the shell's localized power-spectrum is given by

$$^{(2)}n_s = -\frac{2}{n}(n-1)(n-2)\,. \quad (36)$$

The interpretation of short-distance correlations in curved spacetime naturally admits an EFT description, in which correlation lengths are endowed with physical significance as probes of local geometric structure. This viewpoint is standard in modern EFT approaches to gravitational interactions, where observables are organized according to a hierarchy of scales (see for example Refs. 6, 7, and 15).

In this language, our analysis corresponds to an **effective near-core expansion**. In the case of the anisotropic shell, the core radius r_c defines a physical cutoff scale that regulates both the classical geometry and the associated quantum fluctuations. The asymptotic regime $r/r_c \ll 1$ therefore captures the leading contribution to the Wightman function in the region where the spacetime is well defined, while remaining insensitive to the unresolved structure of the naked singularity beyond the shell.

An analogous reasoning applies to the void solution, where the absence of a curvature singularity implies that short-distance correlations are locally Minkowskian up to a conformal modulation. In both cases, the coincidence limit of the Wightman function is taken only after restricting to a controlled regime in which the separation of scales required by EFT is satisfied. This procedure isolates the universal, geometry-induced contributions to vacuum fluctuations while remaining agnostic about the ultraviolet completion of the background.

IV. STRING-INSPIRED INTERPRETATION AND DIMENSIONAL PHASE STRUCTURE

The solutions presented in Secs. I.A and I.B admit a natural interpretation as higher-dimensional, defect-like relics within effective gravitational theories. The discussion in this section is intended to provide physical intuition and to highlight structural parallels with string-inspired and topological scenarios, rather than to propose a direct embedding into a fundamental string-theoretic construction.

Phenomenological interpretation

The anisotropic void and shell solutions can be interpreted as two complementary realizations of extended, defect-like configurations in higher-dimensional effective gravity. In both cases, the gravitational response is not governed by a localized mass monopole but instead arises from anisotropic stress distributions that imprint global geometric features on the spacetime.

The void corresponds to a stress-dominated configuration in which the gravitational field is supported primarily by isotropic pressure and shear stresses, with the energy density becoming subdominant and vanishing identically in the critical case $n = 1$. The shear stress structure acts as an effective tension, producing a repulsive gravitational response and a systematic defocusing of null geodesics. As a result, the spacetime exhibits asymptotically conical behavior, with lensing signatures that saturate to a constant deficit angle. The void therefore behaves as a smooth, extended defect whose influence is encoded in global geometry rather than local curvature, closely analogous to tension-supported relics such as global monopoles or cosmic strings.

The shell, by contrast, localizes anisotropic stress near a finite core scale, generating directional pressure gradients which are not able to regulate a central curvature singularity. Although the shell does not induce a mass-like monopole term, its non–vanishing shear stress gives rise to observable lensing and semiclassical polarization effects. Notably, these effects persist even in the critical case $n = 1$, where the energy density vanishes while anisotropic stresses remain finite. In this limit, the shell retains a finite deflection angle, indicating that its gravitational imprint is controlled by global structure rather than by local curvature dynamics. This behavior closely parallels that of cosmic strings, whose gravitational influence arises from a deficit angle rather than from Newtonian attraction.

Taken together, the void and shell illustrate how extended stress configurations in higher-dimensional gravity give rise to distinct universality classes of gravitational phenomena. The void realizes a smooth, stress-dominated defect with topological asymptotics, while the shell represents an unregulated, anisotropic core whose influence is controlled by a physical cutoff scale. Both examples demonstrate that gravitational repulsion, lensing, and semiclassical effects can emerge from stress anisotropy and global geometry alone, without invoking a cosmological constant or fundamental string degrees of freedom. From this perspective, the void and shell provide effective geometric descriptions of string-like or defect-like relics in higher-dimensional gravitational theories.

Dimensional phase structure

The matter content given in eqs. (6) and (11) exhibits a pronounced dependence of the effective stresses and energy density on the number of spatial dimensions n. For $n > 1$, the energy density is generically negative, signaling a tension-dominated regime. In this higher-dimensional phase, the two solutions display clearly distinct behavior: the void produces a saturated lensing signal characteristic of a topological defect, whereas the shell's deflection angle decays in the weak-field limit, reflecting the absence of a long-range monopole contribution.

At the critical value $n = 1$, the void and shell geometries coincide and both exhibit a saturated deflection angle. This behavior reflects the purely topological nature of gravity in two spacetime dimensions, where curvature does not mediate long-range forces and Einstein gravity possesses no local propagating degrees of freedom. In this regime, gravitational effects are instead sourced by anisotropic stress rather than by energy density. Indeed, while the energy density vanishes identically at $n = 1$, anisotropic stresses persist through a nonzero shear component of the stress–energy tensor. This residual shear governs the remaining gravitational response, including the finite lensing observed for both configurations.

For the formally continued case $n < 1$, the energy density becomes positive. Although this regime does not admit a direct geometric interpretation, it can be accessed through analytic continuation in analogy with dimensional regularization (Ref. 10). From this viewpoint, the sign change of the energy density and pressure reflects a transition in the effective stress–energy description rather than the existence of a spacetime with fractional dimension. Similar dimension-dependent stress–energy behavior is well known in the gravitational description of extended objects and topological defects, where tension rather than energy density controls the large-scale geometry (see Refs. 1–3).

Taken together, these results identify a critical dimensional crossover at $n = 1$. Above this value, the solutions behave as tension-dominated, string-like defects with dimension-dependent lensing properties, while at $n = 1$ the gravitational response becomes purely topological and is entirely governed by anisotropic stress. We refer to this behavior as a *dimension-driven phase structure*, by analogy with dimensional crossovers in statistical and quantum field systems, where qualitative properties of the theory are controlled by the spacetime dimensionality (Ref. 19).

At the semiclassical level, both models present a spectral tilt whose behavior is controlled by the codimension of the defect, $n_s \propto -(n-1)$. For $n > 1$, long-wavelength modes dominate and vacuum fluctuations are enhanced near the core, yielding a red-tilted power spectrum. Closely related infrared enhancements appear in Casimir-like polarizations and vacuum rearrangements around topological defects and extended objects (Ref. 12) further supporting the interpretation of the solutions as extended relics.

In contrast, for topological $(1+1)$ gravity no preferred scale, enhancement, or suppression of wavelengths arises. Correlation functions reduce to their flat-space form due to the absence of transverse degrees of freedom, as is characteristic of conformally invariant two-dimensional theories and is naturally captured in the Polyakov formulation of quantum gravity and string theory (Refs. 5 and 18). The case $n = 1$ therefore finds a direct analogue in the fixed-point behavior of critical phenomena in two dimensions, extensively studied within conformal field theory (Ref. 20).

Conversely, the regime $n < 1$ corresponds to a blue-tilted, UV-dominated sector where short-distance fluctuations prevail. This motivates its interpretation within an effective field theory framework as a formal diagnostic tool for identifying ultraviolet sensitivities and divergences rather than as a physical spacetime configuration (see Refs. 6 and 7). A more detailed topological or dynamical classification of this transition lies beyond the scope of the present work and is left for future investigation.

V. FINAL DISCUSSION

Mathematical observations

Both the void and shell solutions are obtained by imposing geometric constraints that reduce the field equations to a Bernoulli-type differential equation. This restriction reflects the high degree of symmetry of the configurations and enforces a logarithmic-derivative structure for the metric potentials, consistent with the scale-insensitive nature of curvature observables.

Despite this reduction, no physical information is lost. The integration constants of the resulting second-order ordinary differential equations encode global properties of the spacetime, including its asymptotic behavior and topological character. Thus, the imposed Bernoulli constraints provide a controlled mechanism for isolating physically relevant global features that are not captured by local curvature invariants alone.

Concluding remarks

In this work, we have presented two exact, static solutions of the higher-dimensional Einstein field equations supported by anisotropic, geometry-sourced matter, and analyzed their physical implications at both the classical and semiclassical levels. Although the void and anisotropic shell are generated through closely related geometric constraints, they realize distinct effective gravitational regimes, exhibiting qualitatively different geodesic, lensing, and quantum responses.

At the classical level, the void produces a saturated deflection angle that approaches a finite asymptotic value with fractional power-law corrections, a characteristic signature of an extended topological defect with finite core size. The anisotropic shell, by contrast, generates a short-ranged lensing signal that decays more rapidly than in Schwarzschild–Tangherlini spacetimes, reflecting the absence of an effective mass monopole despite the presence of strong curvature and stress near the core. In both cases, the lensing behavior is controlled primarily by global geometric structure rather than by asymptotic Newtonian tails.

At the semiclassical level, quantizing a massless scalar field on these backgrounds reveals a dimension-dependent modification of vacuum fluctuations. In the regulated coincidence limit taken near the characteristic scale r_c, the Wightman function exhibits an effective spectral index whose dependence on the number of spatial dimensions tracks the anisotropic stress content of the geometry. This indicates a scale-dependent vacuum polarization that remains sensitive to the core structure while suppressing direct sensitivity to curvature singularities at the level of observables.

Taken together, the effective energy density profiles, gravitational lensing, and semiclassical correlators consistently point to a dimension-driven phase structure, with a critical crossover at $n = 1$. Notably, this crossover is also suggested by the vanishing of the Ricci scalar, demonstrating that the qualitative phenomenology is robust and governed by codimension and symmetry rather than by the detailed microphysical origin of the matter sector.

The dimensional crossover identified here can be interpreted as signaling the emergence of an effectively topological gravitational regime. As the number of transverse dimensions is reduced, local curvature dynamics become increasingly subdominant, and physical observables, such as lensing saturation, tension-dominated stress, and scale-invariant correlators, become controlled by global geometric features. The case $n = 1$ marks a limiting regime in which deficit angles, saturated deflection, and vacuum structure dominate the phenomenology, closely paralleling aspects of topological gravity in low dimensions.

A more complete treatment of semiclassical backreaction, including a fully renormalized stress–energy tensor and extensions to interacting quantum fields, is deferred to future work.

ACKNOWLEDGMENTS

The author carried out this work as an independent researcher, without institutional affiliation or external funding.

Appendix A: Curvature calculations

The Christoffel symbols of the Newtonian metric (1) are

$$
\begin{aligned}
\Gamma^0{}_{00} &= \frac{1}{2}\frac{\dot{\Psi}}{\Psi}\,, \\
\Gamma^0{}_{0i} &= \frac{1}{2}\frac{\partial_i\Psi}{\Psi}\,, \\
\Gamma^0{}_{ij} &= \frac{1}{2}\frac{\dot{\Phi}}{\Psi}\delta_{ij}\,, \\
\Gamma^i{}_{00} &= \frac{1}{2}\frac{\delta^{ik}\partial_k\Psi}{\Phi}\,, \\
\Gamma^i{}_{j0} &= \frac{1}{2}\frac{\dot{\Phi}}{\Phi}\delta^i{}_j\,, \\
\Gamma^i{}_{jk} &= \frac{1}{2}\left(\delta^i{}_k\frac{\partial_j\Phi}{\Phi} + \delta^i{}_j\frac{\partial_k\Phi}{\Phi} - \delta_{jk}\delta^{il}\frac{\partial_l\Phi}{\Phi}\right).
\end{aligned}
\tag{A1}
$$

From (A1) the Ricci components are

$$
\begin{aligned}
R_{00} = -\frac{n}{2}\frac{\ddot{\Phi}}{\Phi} &+ \left(\frac{n}{2} - \frac{n^2}{4}\right)\frac{\dot{\Phi}^2}{\Phi^2} + \frac{n}{4}\frac{\dot{\Phi}\dot{\Psi}}{\Phi\Psi} \\
&+ \left(\frac{n}{2} - \frac{3}{4}\right)\frac{\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\Psi}{\Phi^2} \\
&+ \frac{\boldsymbol{\nabla}^2\Psi}{2\Phi} - \frac{|\boldsymbol{\nabla}\Psi|^2}{4\Phi\Psi}\,,
\end{aligned}
\tag{A2a}
$$

$$
\begin{aligned}
R_{0i} = \left(-\frac{n^2}{4} + n - \frac{3}{8}\right)\frac{\dot{\Phi}\partial_i\Phi}{\Phi^2} \\
+ \left(\frac{1}{2} - \frac{n}{2}\right)\frac{\partial_i\dot{\Phi}}{\Phi} \\
\left(\frac{n}{4} - \frac{1}{4}\right)\frac{\dot{\Phi}\partial_i\Psi}{\Phi\Psi}\,,
\end{aligned}
\tag{A2b}
$$

$$
\begin{aligned}
R_{ij} = &\left[\frac{1}{2}\frac{\ddot{\Phi}}{\Psi} + \left(\frac{n}{4} - \frac{1}{2}\right)\frac{\dot{\Phi}^2}{\Phi\Psi} + \frac{1}{4}\frac{\dot{\Phi}\dot{\Psi}}{\Psi^2}\right]\delta_{ij} \\
&+ \left[-\frac{1}{2}\frac{\boldsymbol{\nabla}^2\Phi}{\Phi} + \left(1 - \frac{n}{4}\right)\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^2} - \frac{\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\Psi}{4\Phi\Psi}\right]\delta_{ij} \\
&- \frac{\partial_i\partial_j\Psi}{2\Psi} + \frac{\partial_i\Psi\partial_j\Psi}{4\Psi^2} + \frac{\partial_i\Phi\partial_j\Psi + \partial_j\Phi\partial_i\Psi}{4\Phi\Psi} \\
&+ \left(1 - \frac{n}{2}\right)\frac{\partial_i\partial_j\Phi}{\Phi} + \frac{3}{4}(n-2)\frac{\partial_i\Phi\partial_j\Phi}{\Phi^2}\,.
\end{aligned}
\tag{A2c}
$$

Likewise, the Ricci scalar is written as

$$
\begin{aligned}
R = n\frac{\ddot{\Phi}}{\Phi\Psi} &- \frac{n}{2}\frac{\dot{\Phi}\dot{\Psi}}{\Phi\Psi^2} + \frac{n(n-2)}{2}\frac{\dot{\Phi}^2}{\Phi^2\Psi} \\
&- \frac{\boldsymbol{\nabla}^2\Psi}{\Phi\Psi} + \frac{2-n}{2}\frac{\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\Psi}{\Phi^2\Psi} + \frac{1}{2}\frac{|\boldsymbol{\nabla}\Psi|^2}{\Phi\Psi^2} \\
&+ (1-n)\frac{\boldsymbol{\nabla}^2\Phi}{\Phi^2} - \frac{(n-1)(n-6)}{4}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^3}\,.
\end{aligned}
\tag{A3}
$$

Ultrastatic frame

Let us assume isotropy and $\Psi = 1$, then only the spatial-spatial component of the Ricci tensor does not vanish. We have

$$R_{ij} = \mathcal{A}(r)\delta_{ij} + \mathcal{B}(r)\frac{x^i x^j}{r^2}\,, \tag{A4a}$$

$$\mathcal{A}(r) := A(r) + \frac{1}{2}\left(\frac{\Phi'}{\Phi}\right)^2, \tag{A4b}$$

$$\mathcal{B}(r) := B(r) + \frac{(n+1)(n-2)}{4}\left(\frac{\Phi'}{\Phi}\right)^2, \tag{A4c}$$

$$f' := \frac{\mathrm{d}f}{\mathrm{d}r}\,, \quad r := |\mathbf{x}|\,,$$

where the second pair of radial functions are defined in the following.

Constraints on anisotropy

In the ultrastatic frame, (A4), the second pair of radial functions are written as

$$A(r) = -\frac{1}{2}\frac{\Phi''}{\Phi} + \left(\frac{3}{2} - n\right)\frac{1}{r}\frac{\Phi'}{\Phi} + \frac{1}{2}\left(1 - \frac{n}{2}\right)\left(\frac{\Phi'}{\Phi}\right)^2, \tag{A5a}$$

$$B(r) = \left(1 - \frac{n}{2}\right)\left(\frac{\Phi''}{\Phi} - \frac{1}{r}\frac{\Phi'}{\Phi}\right) - \left(\frac{n}{2} - 1\right)^2\left(\frac{\Phi'}{\Phi}\right)^2. \tag{A5b}$$

Thus, by taking them as constraints we will be able to manage the intensity of isotropic or anisotropic stress coming from the geometry.

Furthermore, the Ricci scalar is

$$R = \frac{n-1}{\Phi}\left[-\boldsymbol{\nabla}^2\Phi + \frac{6-n}{4}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi}\right]. \tag{A6}$$

Killing equations

We write the coupled set of linear-PDF that describe the symmetries of the ultrastatic Newtonian spacetime, when $\Psi = 1$,

$$\mathfrak{L}_\xi g_{00} = -2\dot{\xi}^0\,, \tag{A7a}$$

$$\mathfrak{L}_\xi g_{0i} = \dot{\xi}_i\Phi(r) - \partial_i\xi^0\,, \tag{A7b}$$

$$\mathfrak{L}_\xi g_{ij} = \frac{\boldsymbol{\xi}\cdot\mathbf{x}}{r}\frac{\Phi'}{\Phi}\delta_{ij} - \frac{\Phi'}{\Phi}\frac{x^i\xi_j + x^j\xi_i}{r} + \partial_i\xi_j + \partial_j\xi_i = 0\,, \tag{A7c}$$

where $\boldsymbol{\xi} = (\xi_i)_i$ and $\xi^\mu = (\xi^0, \xi^k)$, $\xi^k = \Phi^{-1}\delta^{kl}\xi_l$.

When solving this set of eqs. the first two conditions make the Killing vectors ξ^0 and ξ_i time-independent.

Conserved quantities

The metric (14) admits energy and angular momentum conservation because of the symmetries described by the Killing vectors. To exploit these symmetries we use spherical coordinates,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \Phi(r)\big(\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2_{n-1}\big)\,. \tag{A8}$$

Then, the Killing vectors associated with energy and angular momentum conservation satisfy

$$2E := -\xi_\mu\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\,, \qquad \xi_\mu = (-1, 0, \cdots, 0)\,, \tag{A9a}$$

$$L := \eta_\mu\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\,, \qquad \eta_\mu = (0, \cdots, 0, r)\,. \tag{A9b}$$

Appendix B: Scaling exponent

Weak field limit

In normalized units, the orbit eqs. are

$$\Delta\phi = 2\int_{s_0}^{\infty}\frac{\mathrm{d}s}{s\sqrt{2\,{}^{(1)}\Phi(s)(\frac{s}{\mathtt{b}/r_c})^2 - 1}}\,, \tag{B1a}$$

$$\left(\frac{\mathtt{b}}{r_c}\right)^2 = 2\,{}^{(1)}\Phi(s_0)\,s_0^2\,, \tag{B1b}$$

where $a = r_c^\alpha[L^\alpha]$ is chosen accordingly to yield consistent units.

By the isotropy of the void, the PPN expansion of ${}^{(1)}\Phi(s)$ can only depend on even powers of s. On the other hand, the deflection integral is dominated by the physics at the turning point s_0. Because the integrand contains a square root singularity at $s = s_0$, the behavior of the metric exactly at that point dictates the bulk of the deflection. This explains the asymptotic behavior of the deflection angle

$$\begin{aligned}\alpha &= \alpha_0 + \mathcal{O}\left(\frac{1}{s_0^2}\right), \\ &= \alpha_0 + \mathcal{O}\Big([\mathtt{b}/r_c]^{-\frac{2n}{n+2}}\Big)\,,\end{aligned} \tag{B2}$$

where

$$(\mathtt{b}/r_c)^2 \sim 2n^{2/n}s_0^{2+4/n}\,, \tag{B3}$$

is obtained from ${}^{(1)}\Phi(s) \;\sim\; n^{2/n}s^{4/n}$ in the weak field limit.

Furthermore, when eqs. (B1) are calculated exactly, we have

$$\begin{aligned}\Delta\phi &\sim 2\int_{s_0}^{\infty}\frac{\mathrm{d}s}{s\sqrt{2\frac{n^{2/n}}{(\mathtt{b}/r_c)^2}\,s^{2+4/n} - 1}}\,, \\ &\sim \frac{2}{2+4/n}\int_0^1 u^{-1/2}(1-u)^{-1/2}\,\mathrm{d}u\,, \\ &\sim \frac{n\pi}{n+2}\,,\end{aligned}$$

where the last integral is a Beta function for $u = (s_0/s)^{2+4/n}$.

Thus, the weak field limit yields a conical defect of

$$\alpha \sim -\frac{2}{n+2}\pi\,, \tag{B4}$$

which is identical to the numerical result $\alpha_0 = -(2/5)\pi$, when $n = 3$.

Log-log fit

The slope (fig. 8) was extracted using a linear regression of $\log(|\alpha - \alpha_0|)$ against $\log(\mathtt{b})$ in the asymptotic regime. The resulting best-fit value is $\gamma = -1.199999$, with a 95% confidence interval $[-1.200000, -1.199998]$, corresponding to an uncertainty $\Delta\gamma \simeq 0.001$. In total agreement with the theoretical value of $\gamma = -6/5$, eq. (B2).

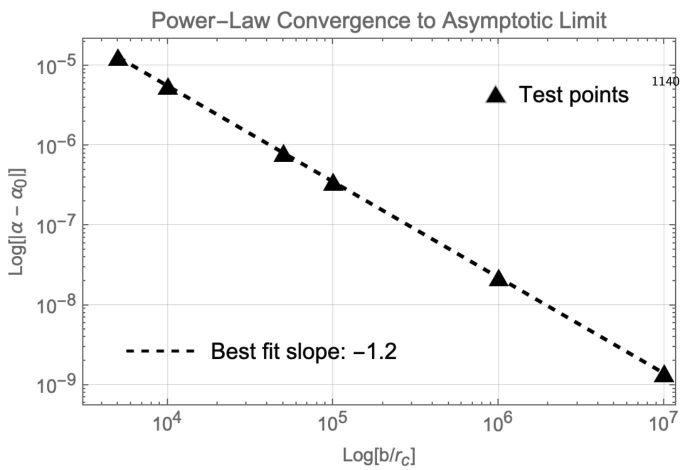

FIG. 8: Log–log fit of the scaling parameter for the conical defect of the void.

Appendix C: CFT in curved spacetime

A scalar field φ interacting with the background geometry has the action

$$S = \int \mathrm{d}\tau \mathrm{d}x^n \frac{\sqrt{-g}}{2}\left(g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi - m^2\varphi^2 - \xi R\varphi^2\right), \tag{C1}$$

from which we obtain the Klein-Gordon (KG) eq.

$$\left(\Box + m^2 + \xi R\right)\varphi = 0\,. \tag{C2}$$

The mass in eq. (C2) introduces a scale that breaks conformal invariance. To study the case when the metric is conformal to Minkowski, let $m = 0$.

The conformal transformation

$$g_{\mu\nu} \to \bar{g}_{\mu\nu} := \Omega^{-2}(x) g_{\mu\nu}\,,$$

allows for the fields in the spacetime $(\mathcal{M}, g)$, $g_{\mu\nu} = \Omega^2(x)\eta_{\mu\nu}$, to be quantized from a **conformal vacuum**.

For instance, the conformal map changes the KG eq. (C2) into

$$\left(\Box + \xi R\right)\varphi = 0 \to \bar{\Box}\bar{\varphi} = 0\,, \tag{C3a}$$

$$\bar{\Box} := \eta^{\mu\nu}\partial_\mu\partial_\nu\,, \tag{C3b}$$

$$\bar{\varphi} := \Omega^{\gamma}(x)\,\varphi\,, \quad \gamma = (n-1)/2\,. \tag{C3c}$$

We notice that this result is independent of the coupling constant ξ.

Consequently, the canonical quantization of the normalized field is given by

$$\hat{\varphi} = \int \mathrm{d}k^n \left(\hat{\mathbf{a}}^\dagger_{\boldsymbol{k}}\varphi^*_{\boldsymbol{k}} + \hat{\mathbf{a}}_{\boldsymbol{k}}\varphi_{\boldsymbol{k}}\right), \tag{C4a}$$

$$\varphi_{\boldsymbol{k}}(x) = \Omega^{\frac{1-n}{2}}(r) u_{\boldsymbol{k}}\,, \tag{C4b}$$

$$u_{\boldsymbol{k}} := \frac{1}{\sqrt{(2\pi)^n\, 2\omega}} e^{-\mathrm{i}\bar{k}\bar{x}}\,, \qquad \bar{k}\bar{x} := -\omega\tau + \boldsymbol{k}\cdot\boldsymbol{x}\,, \tag{C4c}$$

$$\left[\hat{\mathbf{a}}^\dagger_{\boldsymbol{k}}, \hat{\mathbf{a}}_{\boldsymbol{k}}\right] = \delta^{(n)}(\boldsymbol{k} - \boldsymbol{k})\,. \tag{C4d}$$

The equal-time Wightman function is then

$$G^+(\mathbf{x}, \mathbf{x}'; \tau) = \int \frac{\mathrm{d}^n \boldsymbol{k}}{2\pi^n} \frac{\Omega^{\frac{1-n}{2}}(\mathbf{x})\,\Omega^{\frac{1-n}{2}}(\mathbf{x}')}{2k}\, e^{-\mathrm{i}\boldsymbol{k}\cdot(\mathbf{x}-\mathbf{x}')}\,. \tag{C5}$$

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[1] A. Vilenkin, "Cosmic strings," Phys. Rev. D **24**, 2082–2089 (1981).
[2] W. A. Hiscock, "Exact gravitational field of a string," Phys. Rev. D **31**, 3288–3290 (1985).
[3] M. Barriola and A. Vilenkin, "Gravitational field of a global monopole," Phys. Rev. Lett. **63**, 341–343 (1989).
[4] H. Cheng and J. Man, "The deflection angle of a gravitational source with a global monopole in the strong field limit," Classical and Quantum Gravity **28**, 015001 (2010).
[5] K. Becker, M. Becker, and J. H. Schwarz, *String Theory and M-Theory: A Modern Introduction* (Cambridge University Press, 2006).
[6] W. D. Goldberger and I. Z. Rothstein, "Effective field theory of gravity for extended objects," Phys. Rev. D **73**, 104029 (2006).
[7] C. P. Burgess, "Introduction to Effective Field Theory," Ann. Rev. Nucl. Part. Sci. **57**, 329–362 (2007), arXiv:hep-th/0701053.
[8] V. P. Frolov and A. Zelnikov, *Introduction to Black Hole Physics* (Oxford University Press, 2011).
[9] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge Monographs on Mathematical Physics (Cambridge University Press, 1982).
[10] G. 't Hooft and M. Veltman, "Regularization and renormalization of gauge fields," Nuclear Physics B **44**, 189–213 (1972).
[11] C. G. Bollini, J. J. Giambiagi, and A. G. Dominguez, "Analytic regularization and the divergences of quantum field theories," Il Nuovo Cimento (1955-1965) **31**, 550–561 (1964).

[12] J. S. Dowker, "Quantum field theory on a cone," Journal of Physics A: Mathematical and General **10**, 115 (1977).

[13] R. Jackiw, "Lower dimensional gravity," Nuclear Physics B **252**, 343–356 (1985).

[14] C. Teitelboim, "Gravitation and hamiltonian structure in two spacetime dimensions," Physics Letters B **126**, 41–45 (1983).

[15] W. D. Goldberger, "Effective Field Theory for Compact Binary Dynamics," arXiv e-prints , arXiv:2212.06677 (2022), arXiv:2212.06677 [hep-th].

[16] K. Nakajima and H. Asada, "Deflection angle of light in an ellis wormhole geometry," Phys. Rev. D **85**, 107501 (2012).

[17] Y. Lu, X.-Y. Pan, M.-Y. Lai, and Q.-h. Wang, "Finite-distance gravitational lensing of a global monopole in a schwarzschild–de sitter spacetime," Phys. Rev. D **112**, 064051 (2025).

[18] A. Polyakov, "Quantum geometry of bosonic strings," Physics Letters B **103**, 207–210 (1981).

[19] J. Cardy, *Scaling and Renormalization in Statistical Physics*, Cambridge Lecture Notes in Physics (Cambridge University Press, 1996).

[20] P. Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*, Graduate Texts in Contemporary Physics (Springer, 1997).